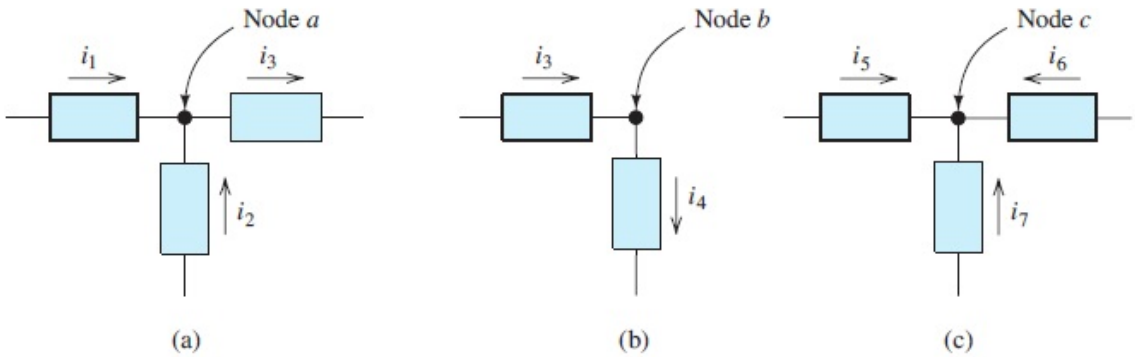
Kirchhoff's Current Law

- A node in an electrical circuit is a point at which two or more circuit elements are joined together. According to Kirchhoff s current law (KCL), the net current entering a node is zero. To compute the net current entering a node, we add the currents entering and subtract the currents leaving.



- For instance, for the above figure, we can write the following

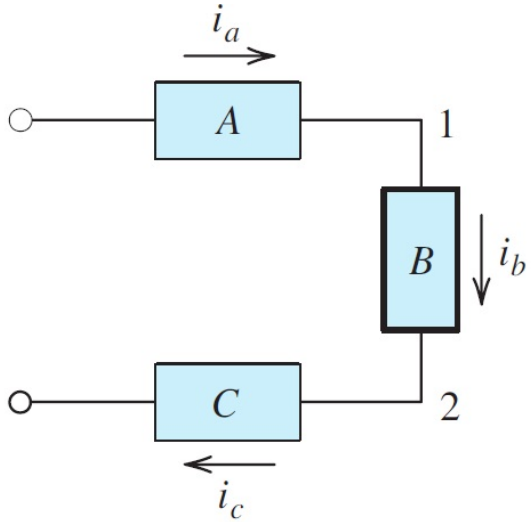
$$\text{Node a: } i_1 + i_2 - i_3 = 0$$

$$\text{Node b: } i_3 - i_4 = 0$$

$$\text{Node c: } i_5 + i_6 + i_7 = 0$$

Series Circuits

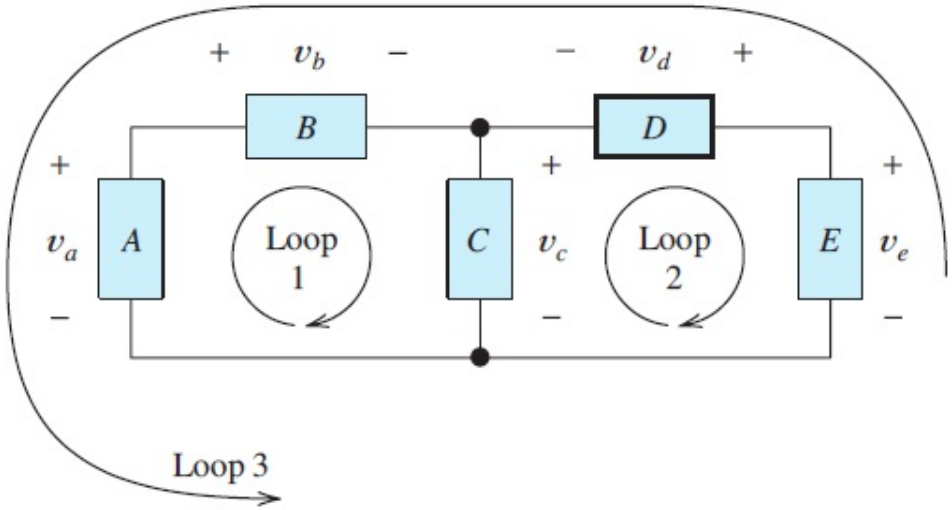
- When elements are connected end to end,we say that they are connected in series. In order for elements A and B to be in series, no other path for current can be connected to the node joining A and B. Therefore, all elements in a series circuit have identical currents.



- For the above circuit, A, B and C are connected in series. Therefore $i_a = i_b = i_c$.

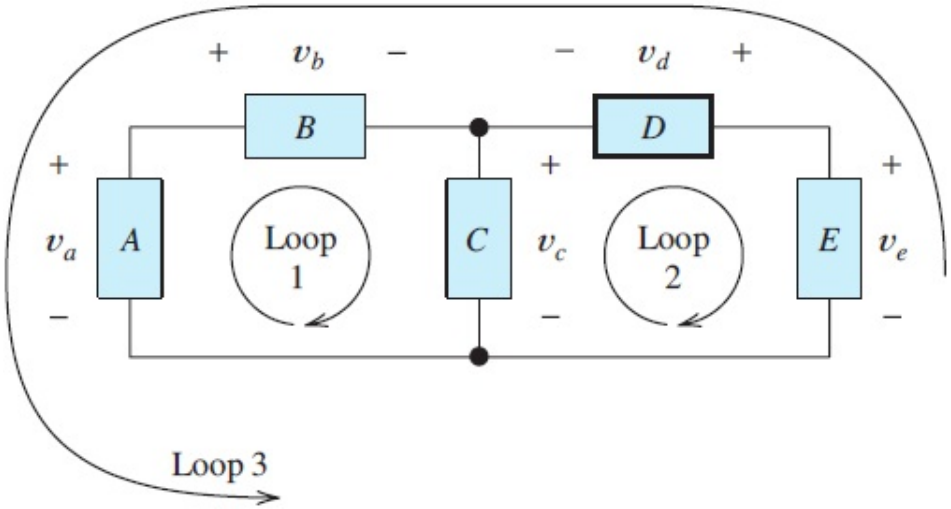
Kirchhoff's Voltage Law

- A loop in an electrical circuit is a closed path starting at a node and proceeding through circuit elements, eventually returning to the starting node. Frequently, several loops can be identified for a given circuit.



- Kirchhoff s voltage law (KVL) states: The algebraic sum of the voltages equals zero for any closed path (loop) in an electrical circuit. In traveling around a loop, we encounter various voltages, some of which carry a positive sign while others carry a negative sign in the algebraic sum. A convenient convention is to use the first polarity mark encountered for each voltage to decide if it should be added or subtracted in the algebraic sum.

Kirchhoff's Voltage Law



- If we go through the voltage from the positive polarity reference to the negative reference, it carries a plus sign. If the polarity marks are encountered in the opposite direction (minus to plus), the voltage carries a negative sign.
- For the above figure, we can write the following

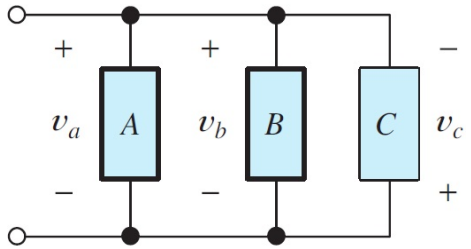
$$\text{Loop 1:} \quad -v_a + v_b + v_c = 0$$

$$\text{Loop 2:} \quad -v_c - v_d + v_e = 0$$

$$\text{Loop 3:} \quad v_a - v_b + v_d - v_e = 0$$

Parallel Circuits

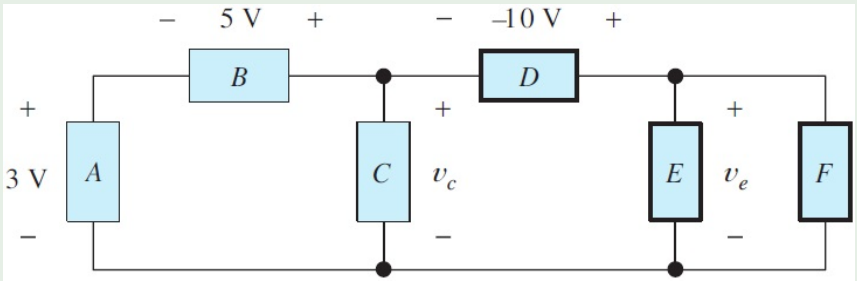
- We say that two circuit elements are connected in parallel if both ends of one element are connected directly (i.e., by conductors) to corresponding ends of the other. The voltages across parallel elements are equal in magnitude and have the same polarity.



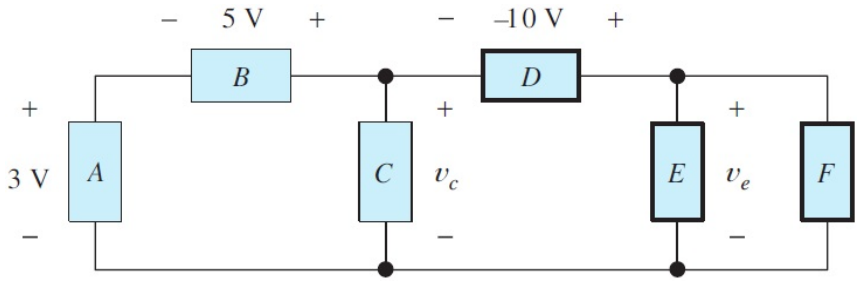
- For the above circuit, we have $v_a = v_b = -v_c$.

Example

For the below electric circuit, find the values of v_c and v_e. Identify elements that are in parallel and that are in series.



Parallel Circuits



- Answer: $v_c = 8V$ and $v_e = -2V$. Elements E and F are in parallel whereas elements A and B are in series.

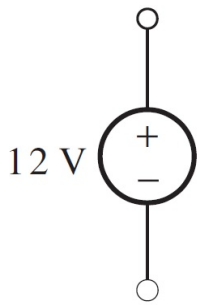
Introduction to Circuit Elements

Conductors

- The voltage between the ends of an ideal conductor is zero regardless of the current flowing through the conductor.

Independent Voltage Sources

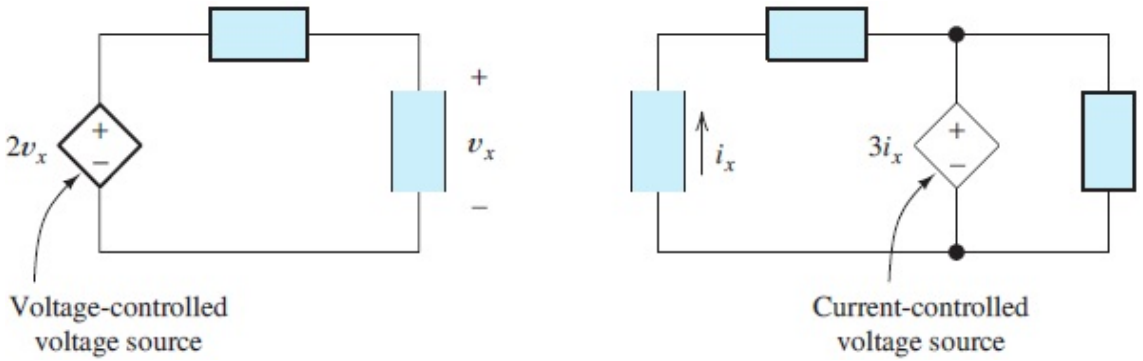
- An ideal independent voltage source maintains a specified voltage across its terminals. The voltage across the source is independent of other elements that are connected to it and of the current flowing through it. We use a circle enclosing the reference polarity marks to represent independent voltage sources. The value of the voltage is indicated alongside the symbol. The voltage can be constant or it can be a function of time.



Introduction to Circuit Elements

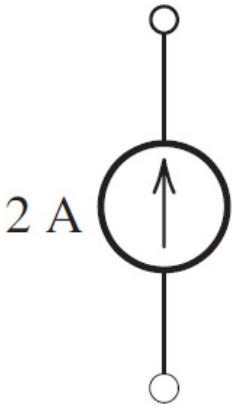
Dependent Voltage Sources

- A dependent or controlled voltage source is similar to an independent source except that the voltage across the source terminals is a function of other voltages or currents in the circuit. Instead of a circle, it is customary to use a diamond to represent controlled sources in circuit diagrams. A voltage-controlled voltage source is a voltage source having a voltage equal to a constant times the voltage across a pair of terminals elsewhere in the network. A current-controlled voltage source is a voltage source having a voltage equal to a constant times the current through some other element in the circuit. The factor multiplying the current is called the gain parameter.



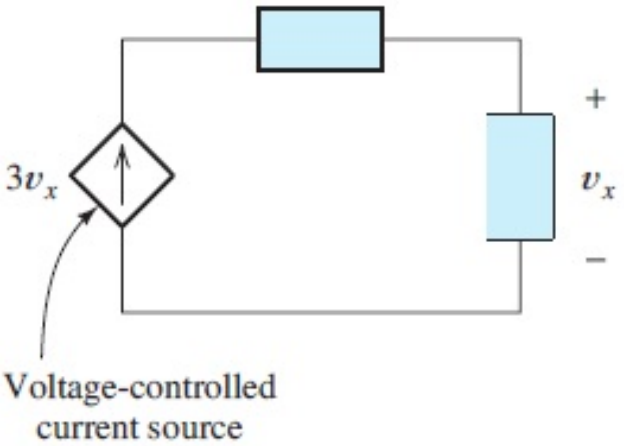
Independent Current Sources

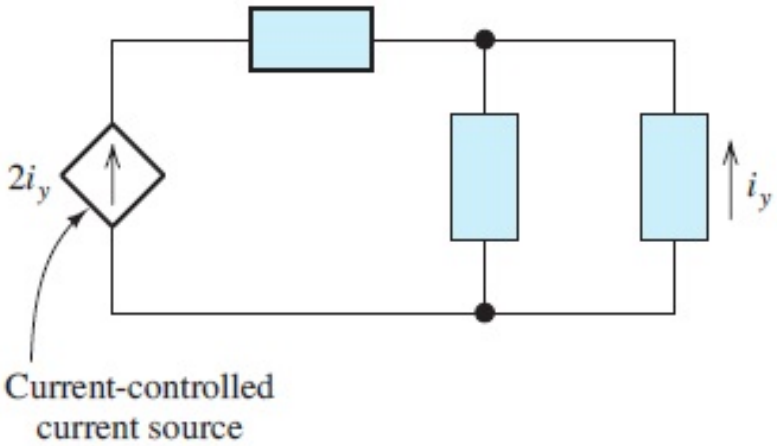
- An ideal independent current source forces a specified current to ow through itself. The symbol for an independent current source is a circle enclosing an arrow that gives the reference direction for the current. The current through an independent current source is independent of the elements connected to it and of the voltage across it.



Dependent Current Sources

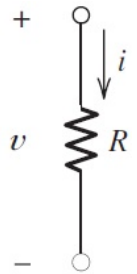
- The current flowing through a dependent current source is determined by a current or voltage elsewhere in the circuit. The symbol is a diamond enclosing an arrow that indicates the reference direction. They may be voltage controlled or current controlled.





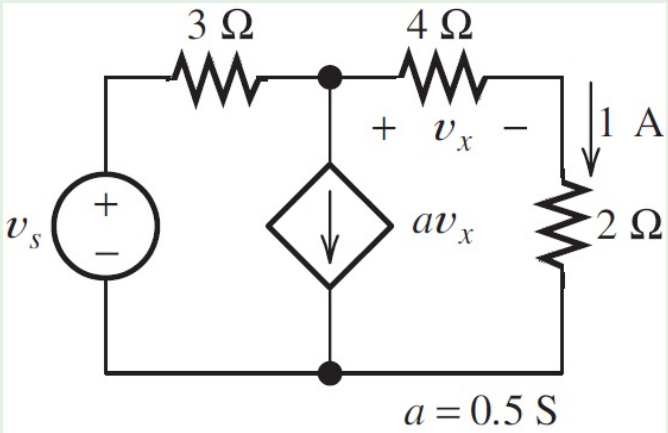
Resistors and Ohm's Law

- The voltage v across an ideal resistor is proportional to the current i through the resistor. The constant of proportionality is the resistance R.
- Ohm's Law: $v = iR$



- The units of resistors are V/A and Ω.

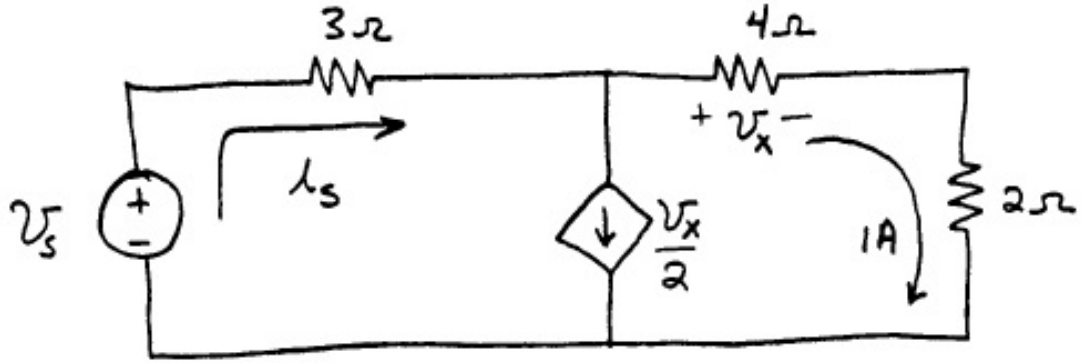
Example



Resistors and Ohm's Law

Solution:

We have a voltage-controlled current source in this circuit.

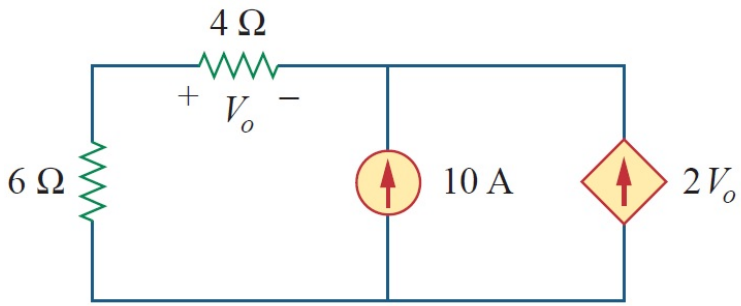


$$v_x = (4\,\Omega) \times (1\,A) = 4\,V \qquad i_s = v_x/2 + 1 = 3\,A$$

Applying KVL around the outside of the circuit, we have:

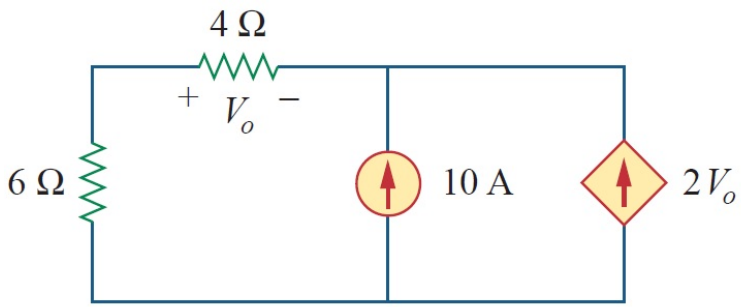
$$v_s = 3i_s + 4 + 2 = 15\,V$$

Resistors and Ohm's Law



Find V_0 in the circuit above and the power dissipated by the controlled source.

Resistors and Ohm's Law



Find V_0 in the circuit above and the power dissipated by the controlled source.

At the node, KCL requires that

$$\frac{v_0}{4} + 10 + 2v_0 = 0 \longrightarrow \quad v_0 = \underline{\mathbf{-4.444V}}$$

The current through the controlled source is

$$i = 2V_0 = -8.888A$$

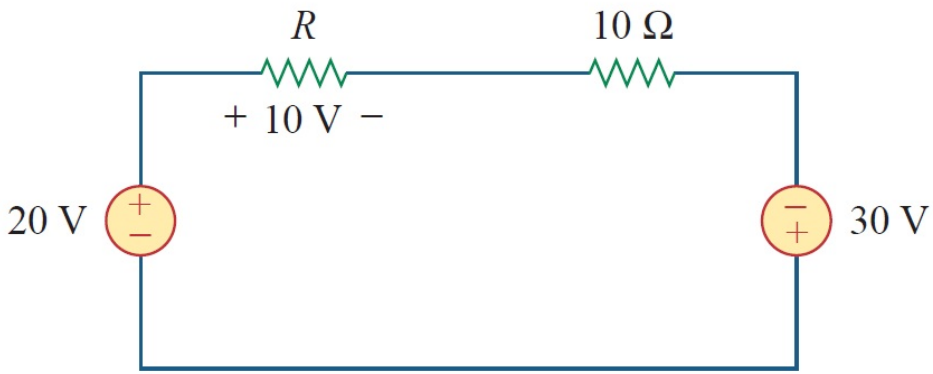
and the voltage across it is

$$v = (6 + 4)\, i_0 \text{ (where } i_0 = v_0/4) = 10\frac{v_0}{4} = -11.111$$

Hence,

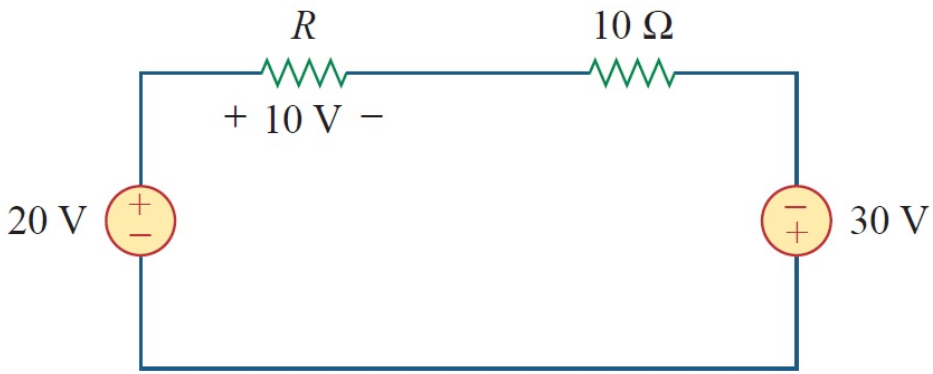
$$p_2\, v_i = (-8.888)(-11.111) = \underline{\mathbf{98.75\ W}}$$

Resistors and Ohm's Law



Find R for the circuit above.

Resistors and Ohm's Law



Find R for the circuit above.

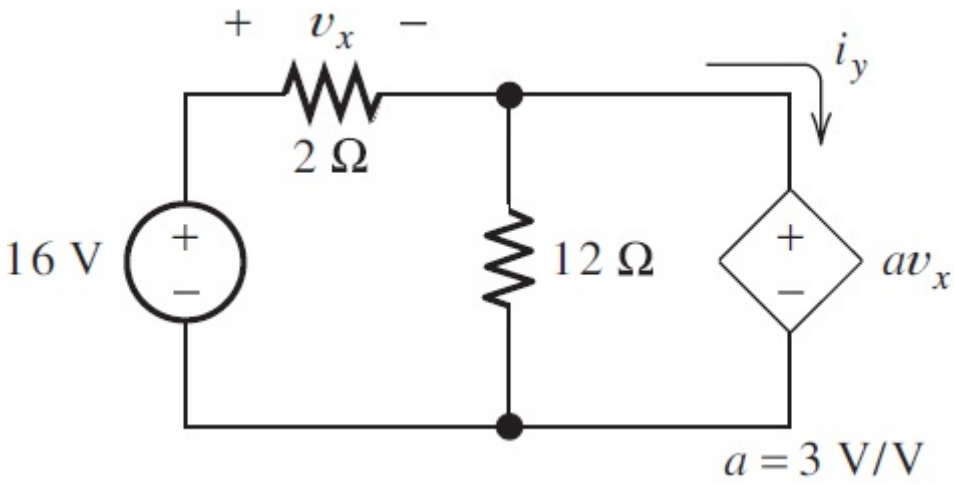
Applying KVL,

$$-20 + 10 + 10I - 30 = 0, \quad I = 4$$

$$10 = RI \longrightarrow R = \frac{10}{I} = \underline{2.5\ \Omega}$$

Resistors and Ohm's Law

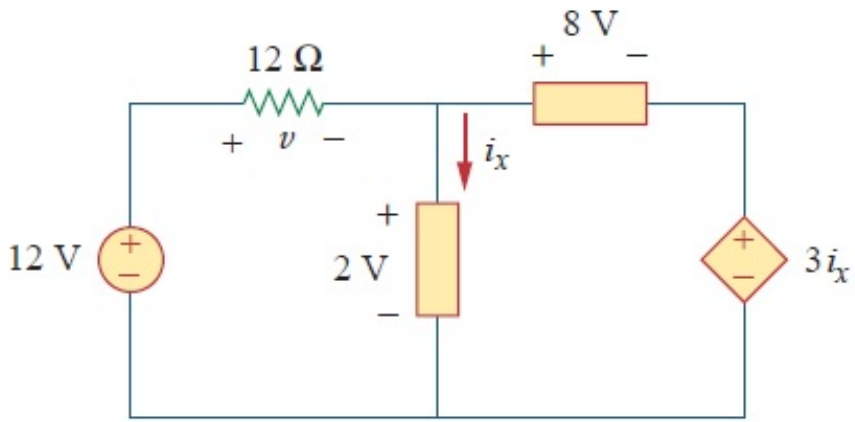
Homework: Consider the circuit given in the below figure. Find i_y and v_x.



Answer: $v_x = 4V$ and $i_y = 1A$.

Resistors and Ohm's Law

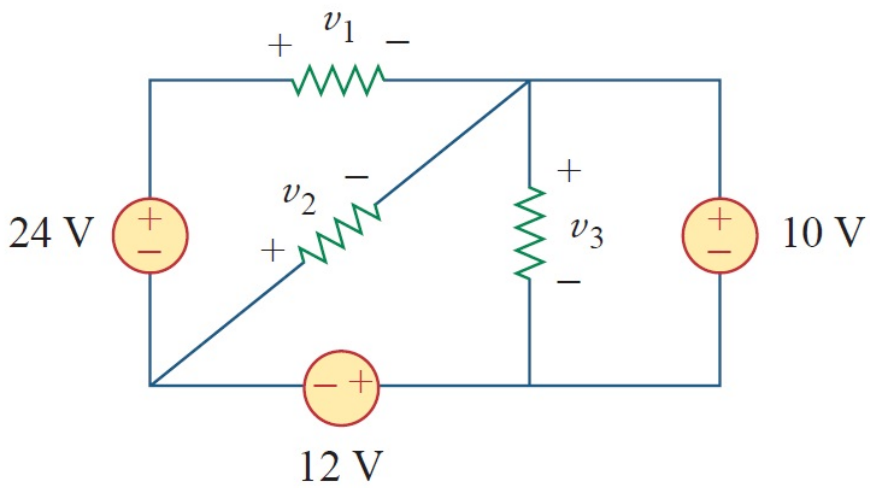
Homework: Consider the circuit given in the below figure. Find v and i_x.



Answer: $v = 10V$ and $i_x = -2A$.

Resistors and Ohm's Law

Homework: Consider the circuit given in the below figure. Find v_1, v_2 and v_3.



Answer: $v_1 = 2V$, $v_2 = -22V$ and $v_3 = 10V$.